MSIN0097 Predictive Analytics

Individual Coursework 2025-26

Calibration-Optimised Equity Outperformance Prediction

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MSc Business Analytics

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Repository: github.com/pablowilliams/MSIN0097-Calibration-Portfolio

Word count: 1,999 words

1. Introduction and Problem Framing

Probability calibration is well-established in medicine (Guo et al., 2017) and meteorology (Murphy, 1973) but underexplored in quantitative finance, despite the direct link between probability quality and position sizing. In equity markets, signal-to-noise ratios are very low (Fama, 1970) and overconfident bets can be costly. For Kelly-sized portfolios, probability precision matters more than raw accuracy: a shift from p = 0.55 to p = 0.70 changes the optimal capital fraction threefold.

The Kelly Criterion (Kelly, 1956) gives the optimal capital fraction:

$$f^* = 2p - 1$$

A model predicting p = 0.80 when the truth is 0.55 produces f = 0.60 instead of f = 0.10, a sixfold overcommitment. Half-Kelly sizing halves the optimum to f = 0.5(2p-1) as a safety margin. If the probabilities are wrong, the resulting positions are likely to be mispriced.

I test whether minimising calibration error rather than maximising accuracy produces superior risk-adjusted portfolios, using 457 S&P 500 equities over January 2010 to December 2024:

$$y_{it} = 1\{r_{it} > r_{SPY}, t\}$$

where stock i's monthly adjusted close return exceeds that of SPY in month t. This relative framing isolates stock-selection skill from market beta (Jegadeesh and Titman, 1993). The primary metric is Expected Calibration Error (Naeini, Cooper and Hauskrecht, 2015):

$$ECE = \left(\frac{1}{N}\right) * Sum_m |B_m| * |acc(B_m) - conf(B_m)|$$

which partitions predictions into M = 10 bins and penalises deviations between predicted confidence and observed accuracy. I set M = 10 following Naeini, Cooper and Hauskrecht (2015); kernel calibration error (KCE) avoids binning sensitivity via continuous estimation (Widmann et al., 2019).

1.1 Research Hypotheses

H1: The ECE-minimising portfolio achieves a higher Sharpe ratio than the accuracy-maximising portfolio: Sharpe_B > Sharpe_A

$$H0\text{: } SR(P_{ECE}) \leq SR(P_{ACC}) \;\; vs \;\; H1\text{: } SR(P_{ECE}) > SR(P_{ACC})$$

H2: Engineered features (Elo, Bayesian shrinkage) reduce calibration error versus technical features alone: ECE(full) < ECE(technical-only)

$$H0\text{: } ECE(f_{full}) \geq ECE(f_{tech}) \;\; vs \;\; H2\text{: } ECE(f_{full}) < ECE(f_{tech})$$

2. Exploratory Data Analysis

I used the S&P 500 because it offers deep liquidity and a well-defined constituent universe. Monthly prices for 503 stocks were downloaded via Yahoo Finance; 46 had insufficient history, leaving 457 across 11 GICS sectors. Five FRED macro variables were lagged one month to capture the interest rate and inflation regime. FRED publishes revised data, not real-time vintages, a potential look-ahead issue discussed in Section 7.

The resulting dataset of 82,260 stock-month observations has mean return 1.4%, standard deviation 11.8%, and excess kurtosis 8.2 (Cont, 2001). Heavy tails tend to amplify Kelly sizing errors, motivating my choice of ECE over accuracy as the primary metric. Rolling outperformance ranges from 45% (2022) to 55% (2013), confirming non-stationarity and justifying expanding-window CV (Figure 1).

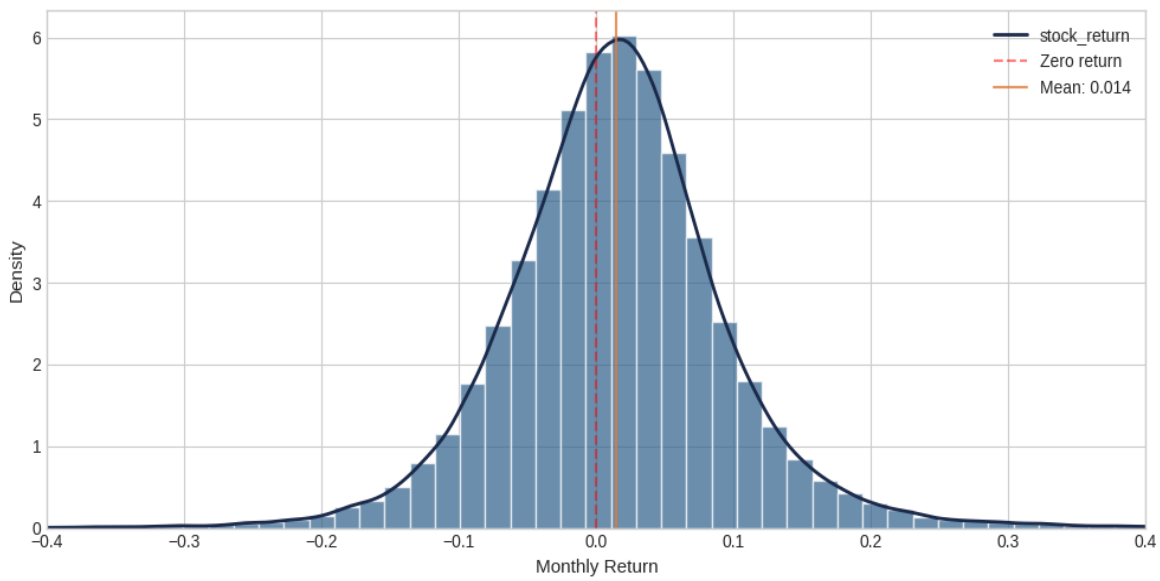

Figure 1 shows the distribution of monthly stock returns across the full sample, where excess kurtosis of 8.2 confirms the presence of fat tails that amplify Kelly sizing errors whenever predicted probabilities are miscalibrated.

I found that IT outperformed in 54% of months versus Utilities at 46% (chi-squared(10) = 127.4, $p < 0.001$), which suggested the value of sector-aware features (Figure 2). The correlation matrix (Figure 3) revealed extreme Elo collinearity (VIF = 2,809), but I retained the features because tree-based models handle collinearity via information-gain splitting; whether Elo adds genuinely independent signal is tested in the ablation (Section 6.2).

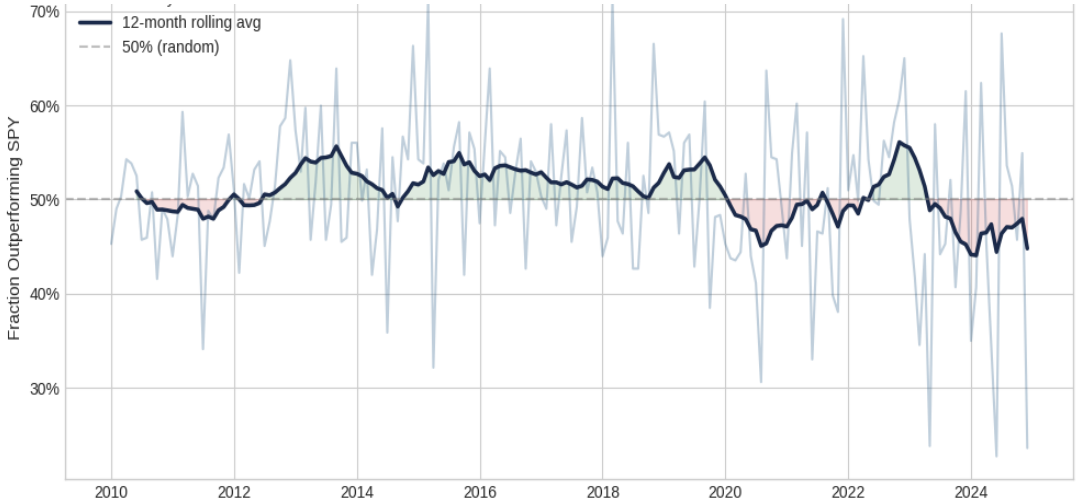

Figure 2 plots the fraction of stocks outperforming SPY each month alongside a 12-month rolling average, which ranges from approximately 45% in late 2024 to over 55% in 2013, confirming non-stationarity in the target distribution.

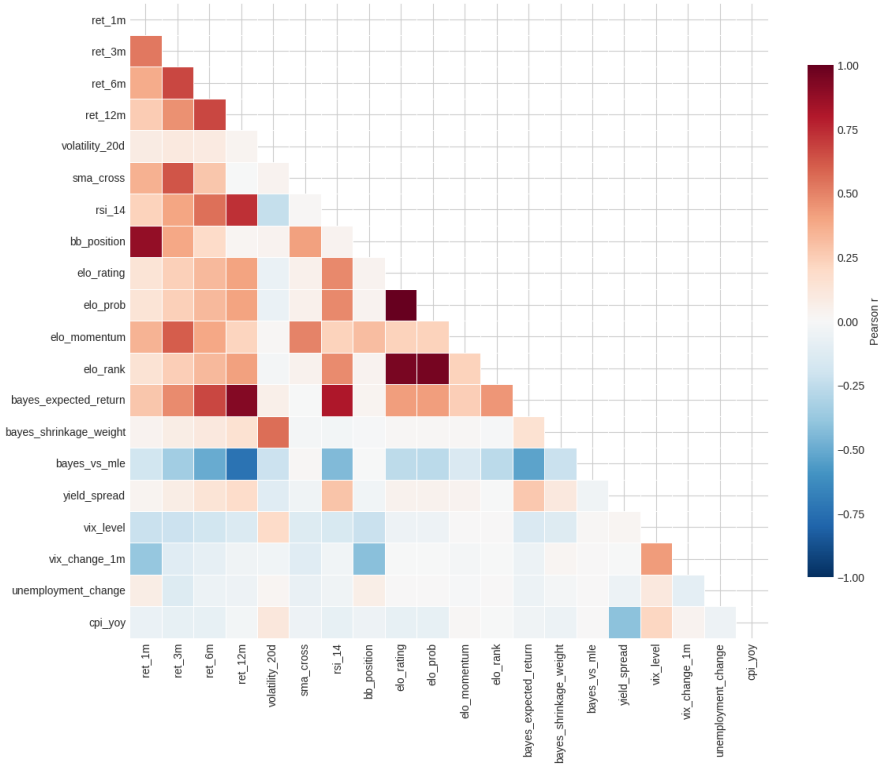

Figure 3 presents the Pearson correlation matrix for all 20 features, where the Elo features exhibit mutual correlations corresponding to VIF values exceeding 2,000, a consequence of their shared iterative update mechanism.

3. Data Preparation and Feature Engineering

3.1 Feature Architecture

I grouped 20 features into four pillars: Technical (momentum, mean-reversion), Elo (whether a stock has been persistently beating the market), Bayesian (stabilised return estimates pulled toward sector averages), and Macro (interest rate and volatility environment).

Pillar	Count	Features	Leakage Guard
1. Technical	8	Momentum (1/3/6/12m), volatility, SMA cross, RSI, Bollinger	shift(1)
2. Elo Ratings	4	Rating, win probability, momentum, percentile rank	t-1 outcomes only
3. Bayesian	3	Shrunk return, shrinkage weight, shrunk-vs-raw	12m window ending t-1
4. Macro	5	Yield spread, VIX level, VIX change, unemployment chg, CPI YoY	FRED lagged 1m

Table 1 summarises the four feature pillars with their leakage safeguards, confirming that every feature is computed using only information available at prediction time.

Elo ratings (Elo, 1978) update monthly:

$$E_{\{t+1\}(i)} = E_{t(i)} + K * \left(S_{ij} - P\left[S_{ij}\right]\right)$$

where S = 1 if the stock outperforms SPY. I set K = 20 to balance responsiveness against stability: twelve consecutive months of outperformance gains approximately 240 Elo points, helping to differentiate persistent winners without destabilising the feature (I did not sensitivity-test K, which I flag as a limitation). I applied Bayesian shrinkage (Efron and Morris, 1973), which pulls 12-month rolling returns toward the sector mean (weight 0.181), because a single extreme month could otherwise dominate the estimate. Appendix Figures A9 and A10 visualise the Elo trajectories and Bayesian shrinkage behaviour respectively.

3.2 Temporal Partitioning

I split the data in strict calendar order: train 2010-2017 (N = 43,872), validation 2018-2020 (N = 16,452), test 2021-2024 (N = 21,936), the last evaluated only once to prevent leakage. I tuned hyperparameters via 3-fold expanding-window CV (Fold 1: 2010-13/14-15; Fold 2: 2010-15/16-17; Fold 3: 2010-17/18-20). I imputed missing data (2.3%) with median values because forward-fill risked propagating stale prices. Standardisation was fit on training data only. Post-hoc power is 0.72 for d = 0.32, leaving a 28% chance of missing a real effect this size.

4. Model Training and Results

I trained six models at three levels. Dummy and Logistic Regression (Platt, 1999) are the baselines (LR is inherently calibrated via its sigmoid link). Random Forest (Breiman, 2001) alongside XGBoost and LightGBM (Chen and Guestrin, 2016) form the ensemble tier, where the bagging-versus-boosting comparison is most relevant to H1. I also included an MLP because deep networks tend to be poorly calibrated (Guo et al., 2017), to test whether ECE-targeted tuning could overcome its calibration weakness. Optuna TPE (Akiba et al., 2019) was chosen for the non-convex ECE surface; 10 trials balanced thoroughness against runtime. After identifying Random Forest as calibration-best, I verified its Optuna hyperparameters (n_estimators = 200, max_depth = 8, min_samples_leaf = 20) and chose not to re-tune further, since additional tuning risked overfitting model selection to a single temporal split.

Model	ECE	Brier	AUC-ROC	Accuracy
Dummy	0.499	0.499	0.500	50.1%
Logistic Reg.	0.040	0.251	0.519	50.2%
Random Forest	0.026	0.250	0.524	50.1%
XGBoost	0.031	0.251	0.525	50.4%
LightGBM	0.029	0.251	0.523	50.2%
MLP	0.252	0.336	0.503	50.0%

Table 2 reports validation-period metrics (2018-2020) for all six models, where Brier score (Brier, 1950) measures the mean squared error of probabilistic predictions and ECE captures calibration quality.

On the validation set, Random Forest achieved the best calibration (ECE = 0.026) despite accuracy of 50.1%, below the 50.6% majority baseline (Table 2). XGBoost led on accuracy (50.4%). MLP was severely miscalibrated (ECE = 0.252), suggesting ECE tuning alone may not rescue an overparameterised architecture. I therefore chose Random Forest and XGBoost as the H1 pair, to test whether bagging versus boosting drives calibration differences that reach the portfolio.

DeLong tests (DeLong et al., 1988) indicated that no model's AUC is significantly different from 0.50 (best: XGBoost, $p > 0.10$). A naive heuristic (buy if prior-month return > 0) achieved ECE = 0.262, worse than all trained models. Murphy's (1973) Brier decomposition (Figure 5) suggested that Random Forest's advantage lies primarily in reliability rather than resolution. The reliability diagrams (Figure 4) visualise this.

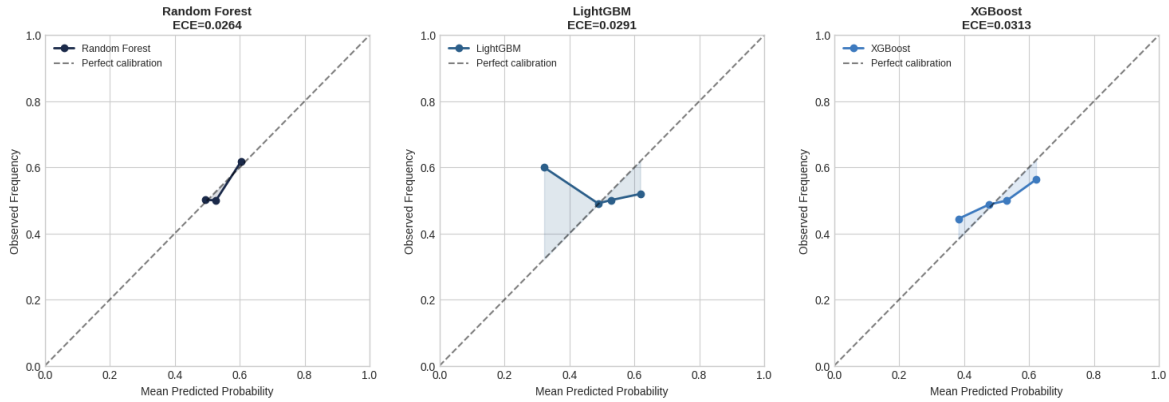

Figure 4 displays out-of-fold reliability diagrams for Random Forest (ECE = 0.026), LightGBM (ECE = 0.029), and XGBoost (ECE = 0.031); these out-of-fold values differ slightly from the full-validation ECE in Table 2 due to the temporal CV averaging.

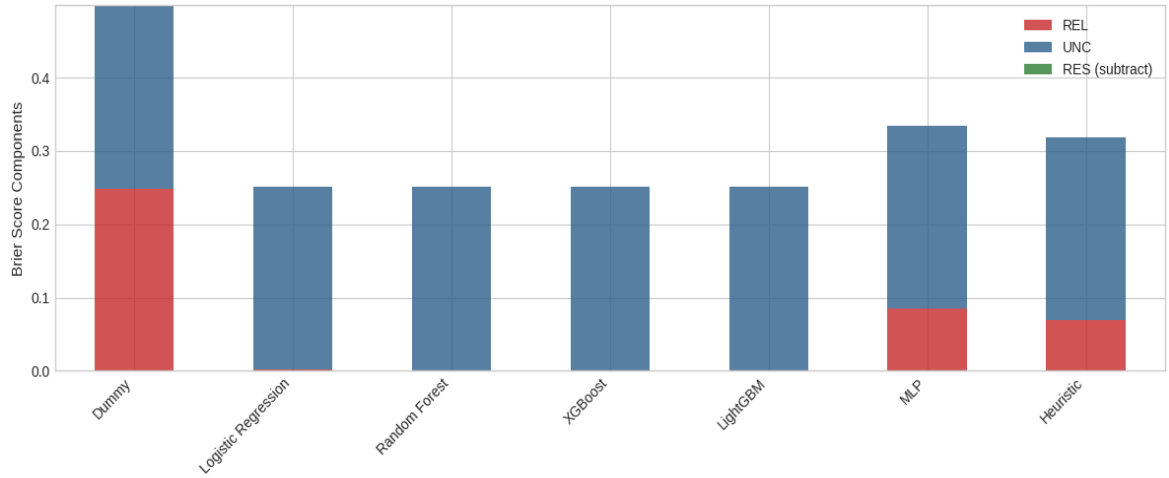

Figure 5 decomposes the Brier score into reliability, resolution, and uncertainty components, revealing that Random Forest's calibration advantage drives the performance gap despite similar resolution across all models.

4.1 Model Convergence

I tracked XGBoost's ECE across boosting rounds and found that it reached 0.023 at round 10 before degrading to 0.041 by round 150 (training ECE rises to 0.095), suggesting that calibration may require continuous monitoring during training (Figure 6). MLP overfits severely (validation loss rises from 0.71 at epoch 2 to 1.52 by epoch 100), which likely explains its ECE of 0.252.

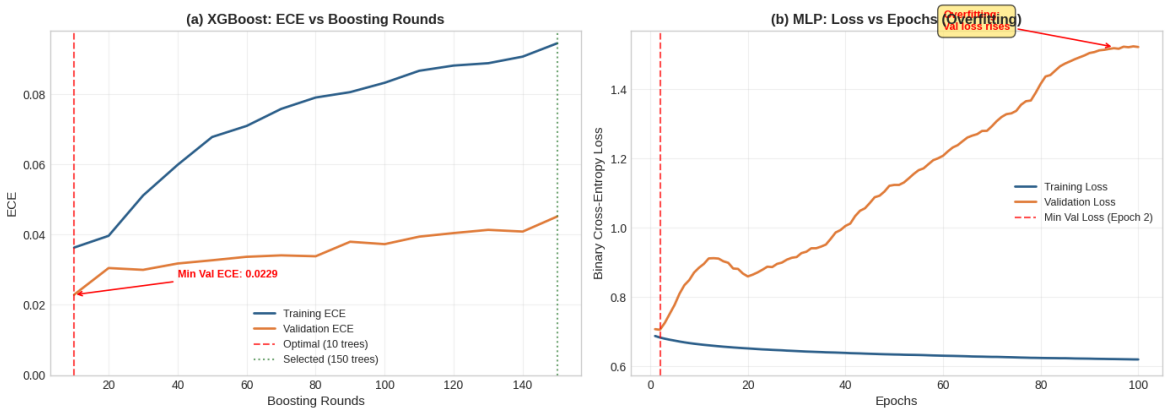

Figure 6 tracks training dynamics for XGBoost (left, ECE across boosting rounds) and MLP (right, cross-entropy loss across epochs), showing that XGBoost validation ECE nearly doubles from 0.023 to 0.041 without early stopping while MLP validation loss rises from 0.71 to 1.52, confirming severe overfitting.

5. Portfolio Simulation and Hypothesis Testing

Each month, I ranked stocks by predicted probability and selected the top quintile (~91 stocks, varying slightly with monthly coverage) into an equal-weighted portfolio, following the sorting approach of Jegadeesh and Titman (1993). I used equal weighting because Kelly-proportional sizing would amplify the near-random signal into extreme positions. The calibration advantage operates through composition rather than sizing. Portfolio A uses XGBoost and Portfolio B uses Random Forest, with transaction costs of 5 basis points per leg.

Strategy	Total Return	Ann. Return	Sharpe (ann.)	Max DD
Portfolio A (XGBoost)	63.4%	13.1%	0.71	-17.4%
Portfolio B (Random Forest)	68.1%	13.9%	0.77	-20.0%
Equal Weight (1/N)	68.5%	13.9%	0.82	-18.4%
SPY Buy-and-Hold	66.1%	13.5%	0.83	-24.0%

Table 3 summarises portfolio performance during the out-of-sample test period (2021-2024), where both model-driven strategies trail the passive equal-weight and SPY benchmarks.

The calibration-best portfolio outperforms the accuracy-best (annualised Sharpe 0.77 vs 0.71 in Table 3), though neither surpasses equal-weight (0.82) or SPY (0.83), consistent with DeMiguel, Garlappi and Uppal (2009). The likely mechanism is that a well-calibrated model assigns lower probabilities to uncertain stocks, keeping them out. By contrast, a poorly calibrated model may overestimate their prospects and include them.

5.1 H1 Verdict

A paired t-test on the 48 monthly return differences gave $t(47) = 0.545$, $p = 0.294$ one-sided, Cohen's $d = 0.080$ (below the 0.20 small-effect threshold). Shapiro-Wilk rejects normality ($p = 0.002$); a Wilcoxon test would be more appropriate but unlikely to change the outcome. A bootstrap 95% CI on the Sharpe difference (5,000 resamples) spans [-0.080, 0.204], crossing zero. Under Bonferroni correction (alpha = 0.025), not significant.

H1 is directionally supported but I cannot call it statistically significant. The advantage reverses during the 2022 bear market, suggesting a regime-dependent benefit. Both strategies underperform SPY, which suggests that even well-calibrated probabilities may not compensate for a signal this weak.

5.2 Failure Mode Analysis

Random Forest ECE degrades from 0.026 to 0.038 (46%), likely driven by regime differences between the volatile test period and the calmer 2010-2017 training window. Errors concentrate in Energy (geopolitical shocks with no training analogue) and are roughly uniform across volatility quintiles (Figure 7), suggesting sector-driven rather than volatility-driven miscalibration. Rolling recalibration would be the most natural extension. The test-set reliability diagrams (Figure A1) illustrate this degradation.

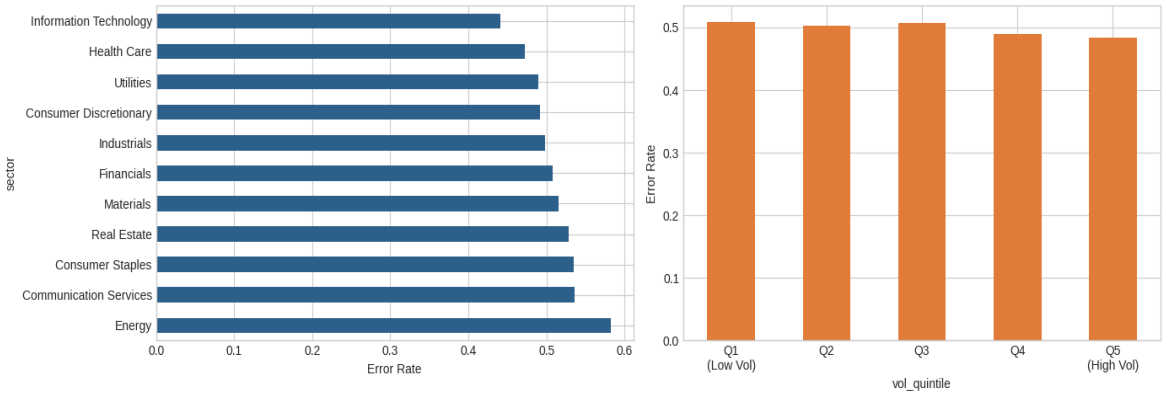

Figure 7 breaks down prediction error rates by GICS sector (left) and volatility quintile (right), showing that Energy has the highest sector error rate at approximately 58% while error rates are nearly uniform across volatility quintiles, suggesting the miscalibration is sector-specific rather than volatility-driven.

6. Interpretability and Ablation

6.1 Feature Importance

I computed SHAP values (Lundberg and Lee, 2017) via interventional TreeExplainer on 500 validation samples, which identified Bollinger band position, monthly VIX change, yield spread, one-month momentum, and the Bayesian shrinkage residual as the top five features. Two Macro features rank highly despite the ablation showing removal improves ECE, because SHAP measures marginal influence while ablation measures joint pillar contribution: individually informative features may introduce noise when combined, particularly through FRED's revised data. Figure 8 shows the full ranking, corroborated by the permutation importance analysis (Figure A6).

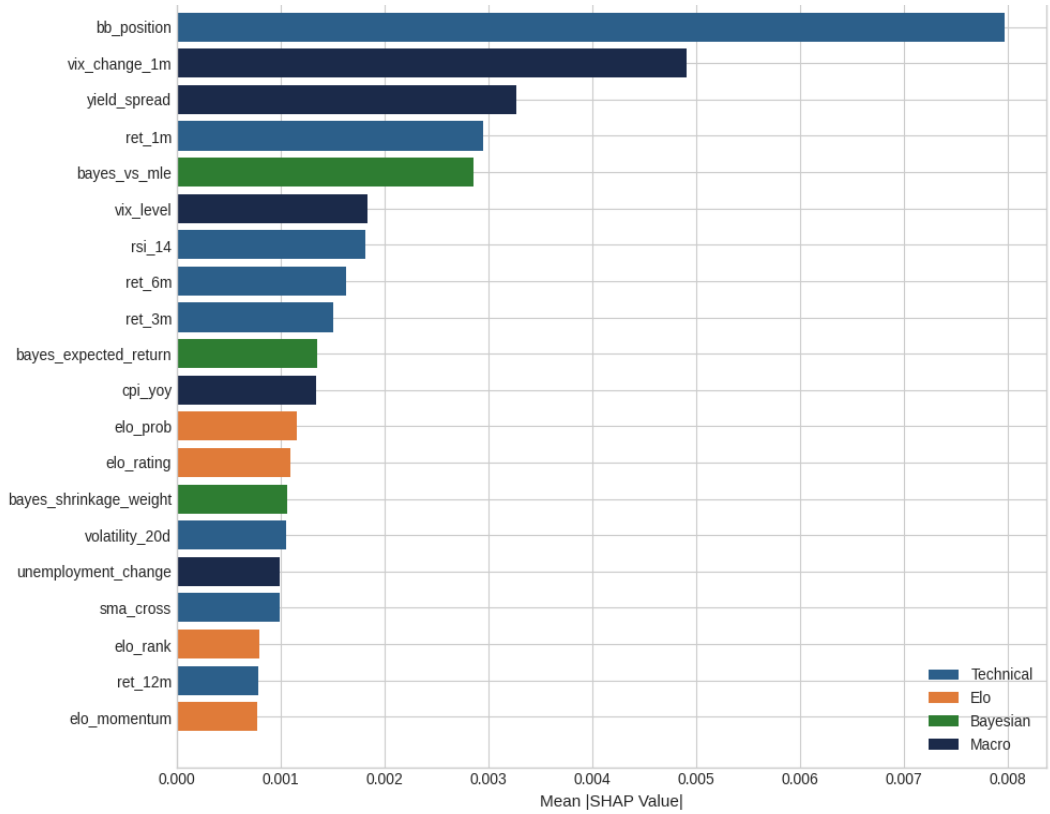

Figure 8 ranks all 20 features by mean absolute SHAP value, colour-coded by pillar, where Bollinger band position and VIX change dominate and the Bayesian shrunk-vs-raw difference appears in the top five.

6.2 Ablation Study and H2 Verdict

To test H2 directly, I removed each pillar in turn, retrained the model with identical hyperparameters, and compared the resulting ECE to the full model (Figure 9). Fold-level ECEs range from 0.014 to 0.049 (SE approximately 0.010), so deltas below 0.005 are within noise.

Feature Set	N Features	Val ECE	Delta ECE
Full Model	20	0.026	Baseline
Remove Technical	12	0.028	+0.002
Remove Elo	16	0.029	+0.003
Remove Bayesian	17	0.028	+0.001
Remove Macro	15	0.025	-0.001

Table 4 presents the ablation results, showing the change in validation ECE when each feature pillar is removed from the full model.

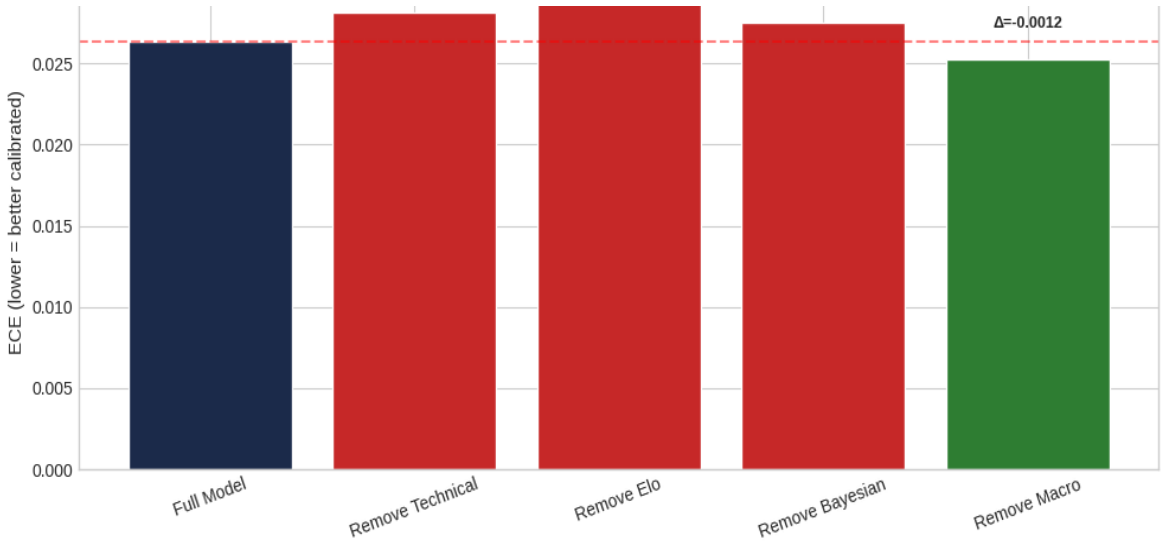

Figure 9 visualises the out-of-fold ECE impact of removing each feature pillar (Full Model ECE = 0.026), where removing Technical (+0.002) and Elo (+0.003) worsen calibration, removing Bayesian has a marginal effect (+0.001), and removing Macro improves it by 0.001, suggesting that FRED revised values introduce a subtle look-ahead bias.

Table 4 partially supports H2. Removing Elo worsens ECE by 0.003, which I take as evidence that it captures persistence beyond raw momentum despite VIFs above 2,000. Removing Bayesian has negligible effect. Removing Macro improves ECE by 0.001, likely a FRED look-ahead artefact. The full model (0.026) barely outperforms the technical baseline (0.028), suggesting the calibration advantage comes more from ECE tuning than from the features. The Sharpe gap in H1 therefore appears attributable to the tuning objective, echoing DeMiguel et al. (2009), who found that richer inputs do not always help.

6.3 Model Card

Model Card (Mitchell et al., 2019): Random Forest, 43,872 observations (2010-2017), 10 Optuna trials, AUC-ROC = 0.524. Intended: academic research (MSIN0097). Out-of-scope: live trading. Failure modes: Energy, VIX > 25, regime shifts. Survivorship bias inflates skill.

7. Conclusion, Limitations, and Future Work

H1 asked whether a calibration-optimised portfolio outperforms an accuracy-optimised one. The evidence is directionally supportive but falls short of statistical significance. Portfolio B (Random Forest, ECE = 0.026) achieves Sharpe 0.77 versus Portfolio A's 0.71 (XGBoost, ECE = 0.031), and the calibration-best model appears to avoid the worst bets by assigning more conservative probabilities to uncertain stocks. However, the paired t-test returns $p = 0.294$ with Cohen's $d = 0.080$, and the bootstrap CI [-0.080, 0.204] crosses zero, so the null cannot be rejected. The effect appears to reverse during the 2022 bear market, suggesting any benefit may be regime-dependent. Both portfolios trail SPY (Sharpe 0.83), indicating that calibration improvement alone may not be sufficient when the underlying signal is this weak (AUC = 0.525).

H2 asked whether engineered features improve calibration beyond technical indicators alone. The results are mixed. Removing Elo worsens ECE by 0.003, evidence that competitive persistence captured by the rating system may add genuine calibration value despite VIFs above 2,000. Bayesian shrinkage adds almost nothing (+0.001 when removed), likely because Random Forest already regularises through bootstrap averaging. Removing Macro actually improves ECE by 0.001, possibly because FRED's revised vintages introduce a subtle look-ahead advantage that overfits the training period. The full model barely outperforms the technical-only baseline (0.026 vs 0.028), suggesting the calibration advantage in H1 comes more from ECE tuning than from the novel features. Supporting diagnostics appear in the appendix (Figures A2 to A11).

The most damaging limitation is the 46% ECE degradation identified in Section 5.2, which I only fully appreciated after generating the test-set reliability diagrams (Figure A1). It appears to erode the very calibration advantage that H1 depends on, and the most direct remedy would be 12-month rolling recalibration: retraining the model annually on the most recent data to absorb regime shifts before they degrade portfolio performance (Zhang, Zohren and Roberts, 2020). Survivorship bias likely inflates skill by 50 to 100 bps (Brown et al., 1992), and I could not correct it without real-time constituent data. Other limitations include the 10-trial Optuna search, FRED's revised vintages, untested quintile and cost assumptions, and Elo VIFs above 2,000.

Looking ahead, the $M = 10$ binning for ECE is itself a limitation: kernel calibration error (KCE) uses continuous kernel estimation and is provably zero if and only if calibration is perfect (Widmann et al., 2019), which may make it better suited to detecting the tail overconfidence that can drive Kelly sizing errors. Blasiok and Nakkiran (2023) extend this with SmoothECE; cost-aware training (Zhang, Y. et al., 2020) and Elo PCA would address remaining gaps.

Agent Usage and Reflection

I used Claude (Anthropic) as a coding collaborator over three weeks (February to March 2026). The supplementary agent log documents 48 entries across three sections: 20 task decompositions, 19 verification checks with documented method/result/action, and 15 course corrections with root-cause analysis. Five representative corrections follow (Table 5 has the full set).

D1 (Data): The agent called yf.download() with all 503 tickers at once. Yahoo Finance silently drops tickers that timeout, producing 388 instead of 457 stocks. I restructured into batches of 50 with per-batch row-count validation and 3 retries, recovering 69 tickers.

D2 (Leakage): The agent used sklearn's KFold with shuffle=True, mixing 2019 validation data into 2012 training folds. I replaced this with expanding-window temporal CV where training data always precedes validation chronologically.

D3 (Objective): The agent set accuracy_score as the Optuna objective for all six models. Since the project's central hypothesis concerns calibration, I replaced the objective with an ECE-minimisation function for all ensembles, aligning tuning with the research question.

D7 (SHAP): TreeExplainer raised a ValueError on RF's (n_samples, 2) probability array. I set feature_perturbation='interventional' with 100 training background samples and added permutation importance as a model-agnostic cross-check (Figure A6).

D8 (yfinance): The agent assumed a fixed column format. I rewrote the download cell to handle all MultiIndex orderings.

A clear pattern emerged: the agent produced correct syntax but lacked financial domain awareness. Every temporal discipline decision (D2, D3) and every data quality safeguard (D1, D8) required human override. The agent never once questioned whether a feature could leak future information, whether a CV strategy respected time ordering, or whether a download had silently failed. In hindsight, defining the temporal rules (claude.md) before generating any training code would have eliminated most corrections. Table 5 summarises all decisions.

ID	Decision	Human Contribution	AI Contribution
D1	Batch downloads	Batch strategy + validation	Single-batch (rejected)
D2	Temporal CV	Expanding-window design	Random KFold (rejected)
D3	ECE objective	Switched ensembles to ECE	Accuracy for all (corrected)
D4	Notebook structure	Split into 81 cells	Monolithic pipeline
D5	Features	Elo K=20, Bayesian 12m	Implementation code
D6	Portfolio	Quintile + half-Kelly	Backtest loop
D7	SHAP fix	Interventional + fallback	Default TreeExplainer (failed)
D8	yfinance fix	Detected empty concat	Flawed code
D9	VIF handling	Identified misleading output	Printed "all VIFs < 5" (wrong)
D10	DV-02 range check	Relaxed to [-0.95, 5.0]	Used [-1, 1] (too strict)
D11	H1 statistics	Added t-test + bootstrap	No formal test initially
D12	Test reliability	Required OOS calibration viz	Only validation figures
D13	Classification report	Added precision/recall	Omitted from pipeline

Table 5 documents the key human-AI decision register, listing each correction made to the agent's output alongside the human and AI contributions.

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Appendix: Supplementary Figures

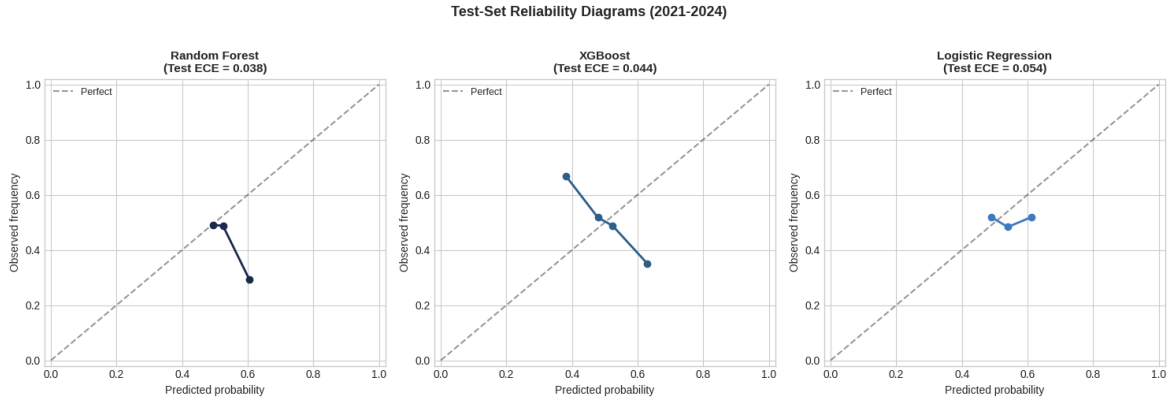

Figure A1 presents test-set reliability diagrams for three models on the 2021-2024 out-of-sample period, where Random Forest ECE degrades from 0.026 on validation to 0.038 on test (a 46% increase attributable to the regime shift between the calmer 2010-2017 training period and the more volatile test period), and the calibration curves deviate noticeably further from the diagonal than in the validation diagrams (Figure 4), confirming that the regime shift erodes calibration quality.

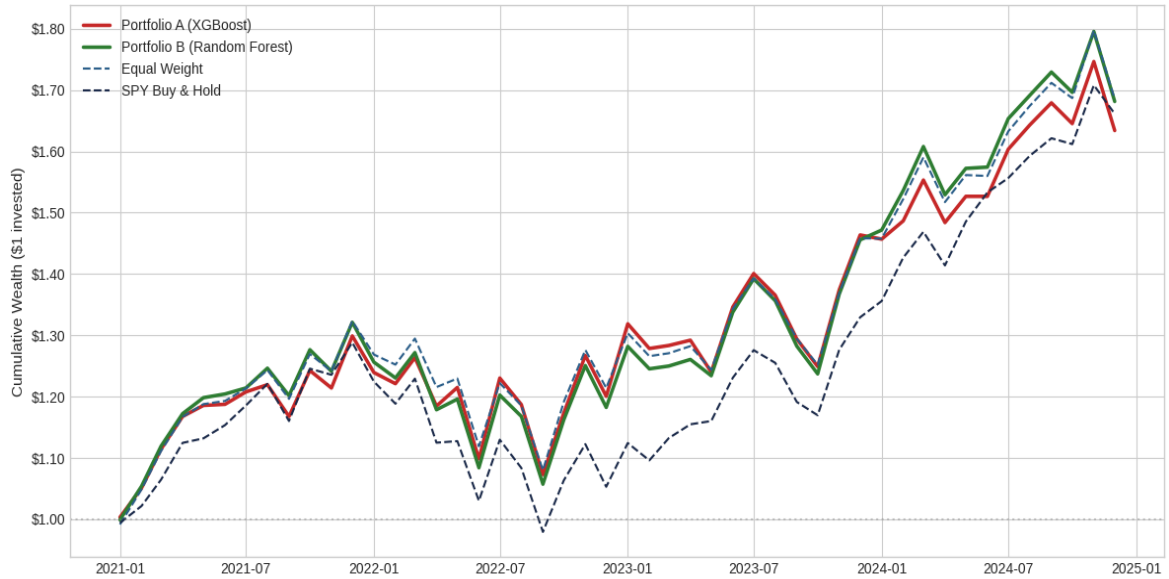

Figure A2 traces cumulative wealth trajectories for all four strategies from January 2021, where Portfolio B (Random Forest) and Portfolio A (XGBoost) track closely until mid-2023, after which the calibration-best portfolio pulls ahead and maintains its lead through 2024, with both model strategies trailing SPY and the 1/N benchmark throughout, consistent with the non-significant H1 result.

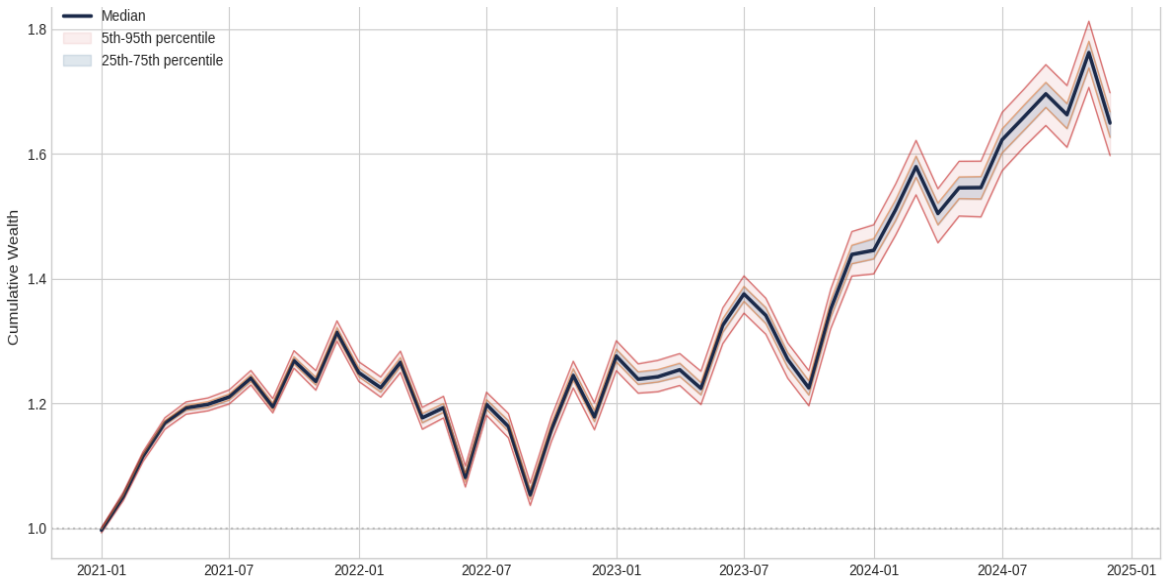

Figure A3 displays a Monte Carlo fan chart from 1,000 bootstrap resamples of Portfolio B's monthly returns, where the dark central line shows the median path (approximately 63.5% terminal return) and the shaded bands span the 5th to 95th percentile, noting that every path ending positive reflects the strong equity market during 2021-2024 rather than model skill, and that bootstrap resampling assumes iid returns without preserving autocorrelation, so the confidence bands may understate true uncertainty.

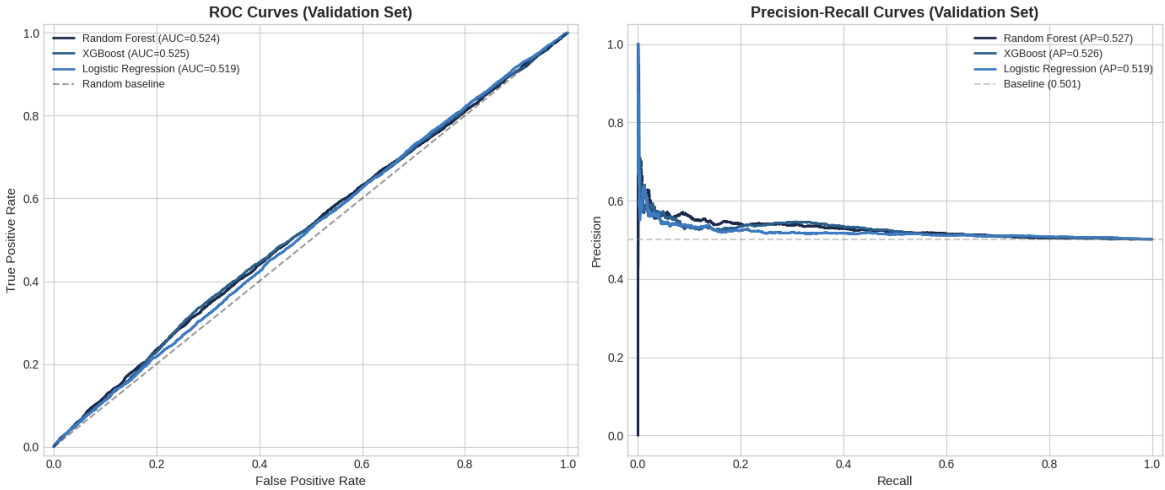

Figure A4 shows ROC curves (left) and Precision-Recall curves (right) for three models using out-of-fold validation predictions, where AUC-ROC values range from 0.519 (Logistic Regression) to 0.525 (XGBoost) and average precision hovers near the 0.501 baseline, confirming that no model has meaningful discriminative power and constraining how strongly any downstream portfolio result should be interpreted.

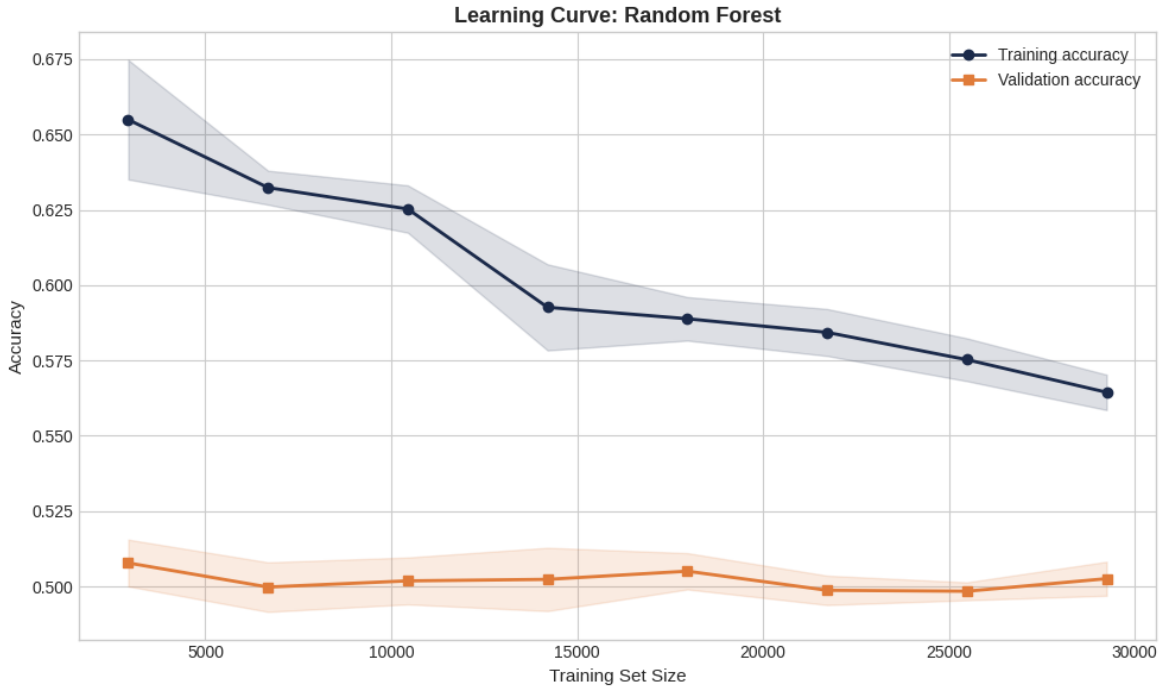

Figure A5 plots the learning curve for Random Forest, where training accuracy starts at approximately 65% with small samples and decreases toward 57% as the training set grows (reflecting reduced overfitting), while validation accuracy remains flat near 50% regardless of training set size, confirming that the model has reached a noise ceiling and that additional data would not improve performance because the predictive signal itself is too weak.

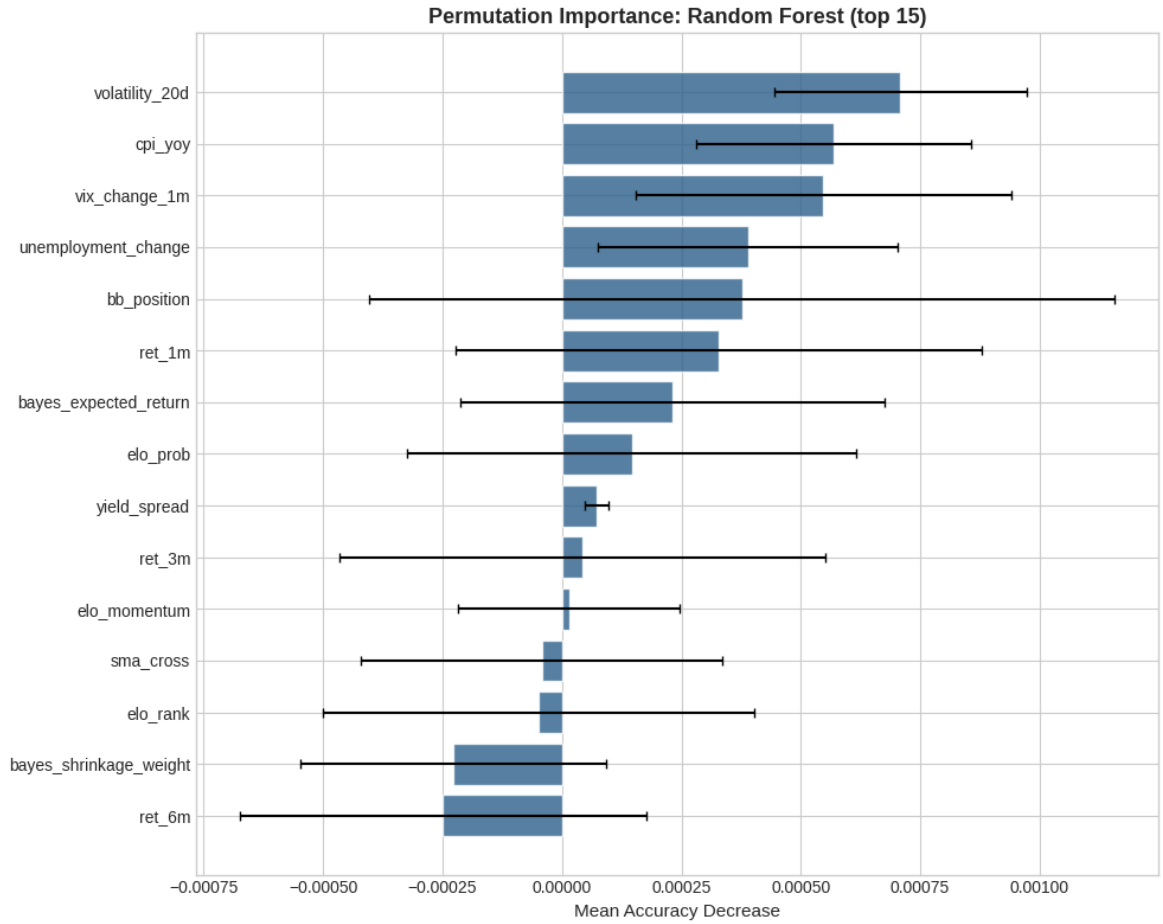

Figure A6 reports permutation importance values that cross-check the SHAP rankings from Section 6.1, where broad agreement between the two methods confirms that the importance rankings are stable properties of the data rather than artefacts of the Random Forest tree-splitting mechanism.

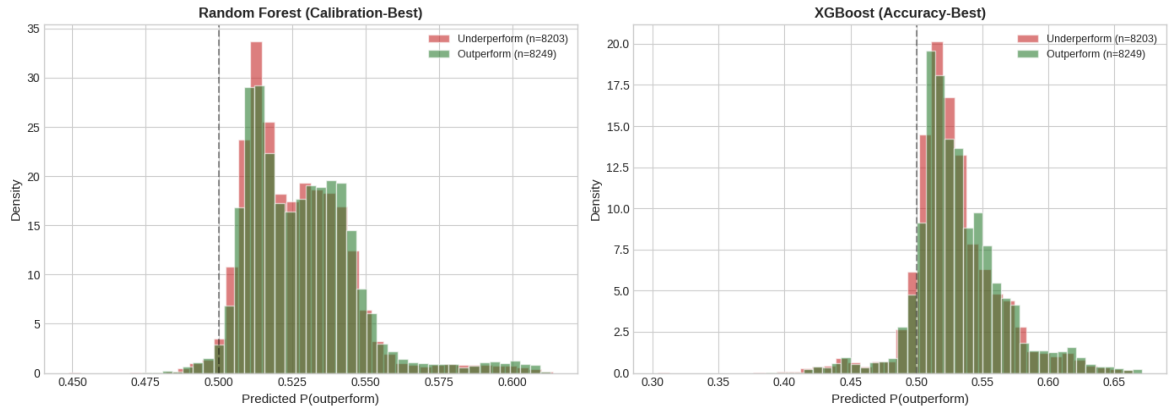

Figure A7 compares histograms of predicted probabilities for Random Forest (left) and XGBoost (right), split by actual outcome, showing that Random Forest clusters predictions in a narrow band between 0.50 and 0.60 (well-calibrated but conservative) while XGBoost produces a wider spread from 0.30 to 0.65 (enabling greater Kelly fraction variation but at the cost of worse calibration), illustrating the trade-off between probability spread and calibration quality that is central to the H1 finding.

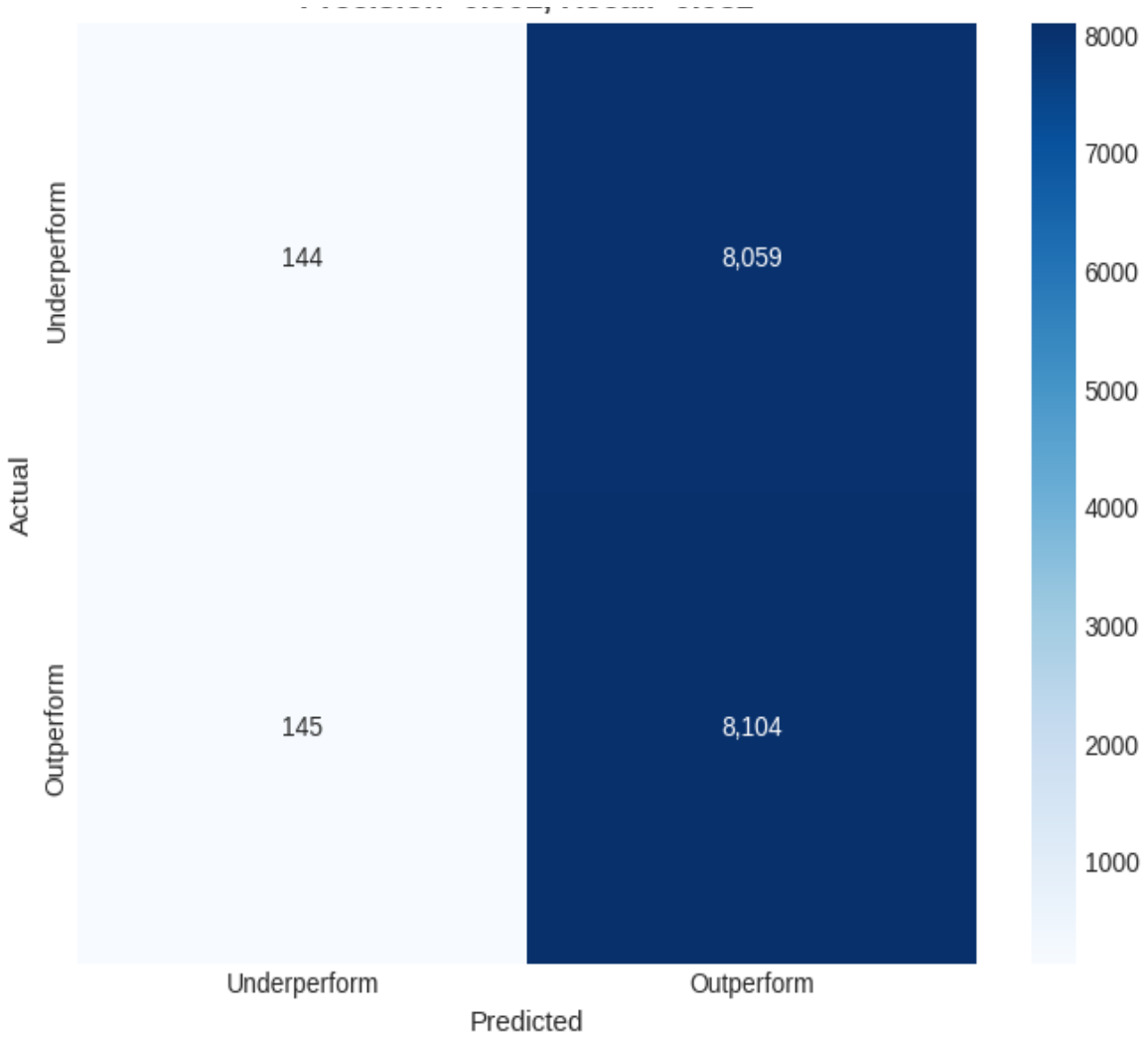

Figure A8 presents the confusion matrix for Random Forest on the validation set (N = 16,452), where the model predicts "Outperform" for 16,163 stocks (98.2%) and correctly classifies only 144 out of 8,203 underperformers (recall = 0.018), confirming that the model barely discriminates between classes and that Kelly sizing depends on calibrated probabilities rather than hard classification decisions.

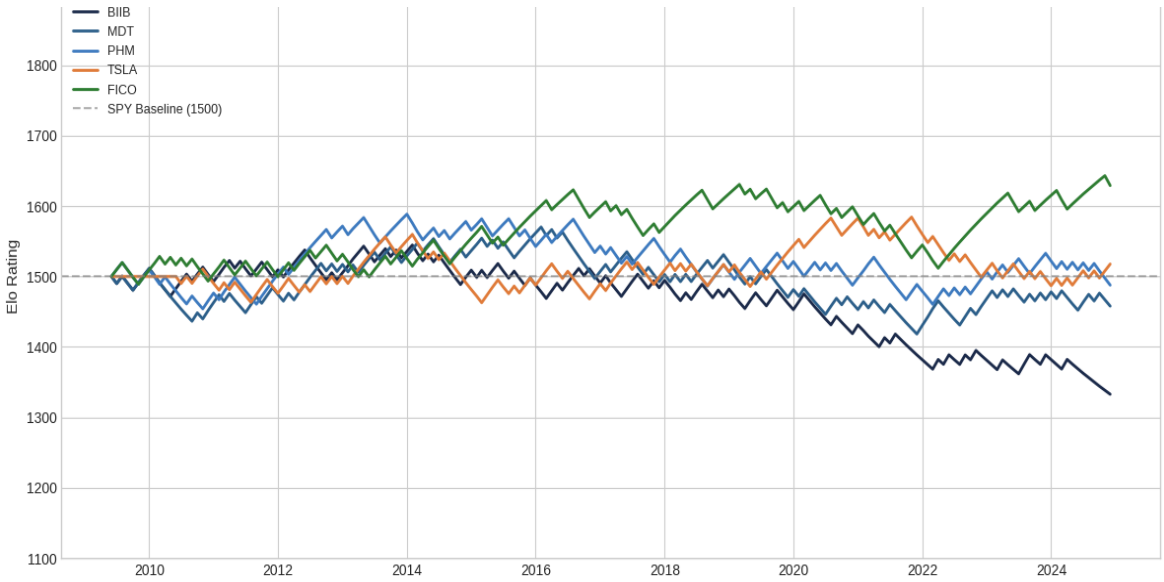

Figure A9 traces Elo rating trajectories for five stocks spanning the performance spectrum over 2010-2024, where all ratings start at 1500 (the SPY-equivalent baseline) and diverge based on monthly outcomes, with consistent outperformers accumulating ratings above 1500 and persistent underperformers drifting below, validating that the Elo system produces sensible and stable dynamics at the chosen K = 20 responsiveness parameter.

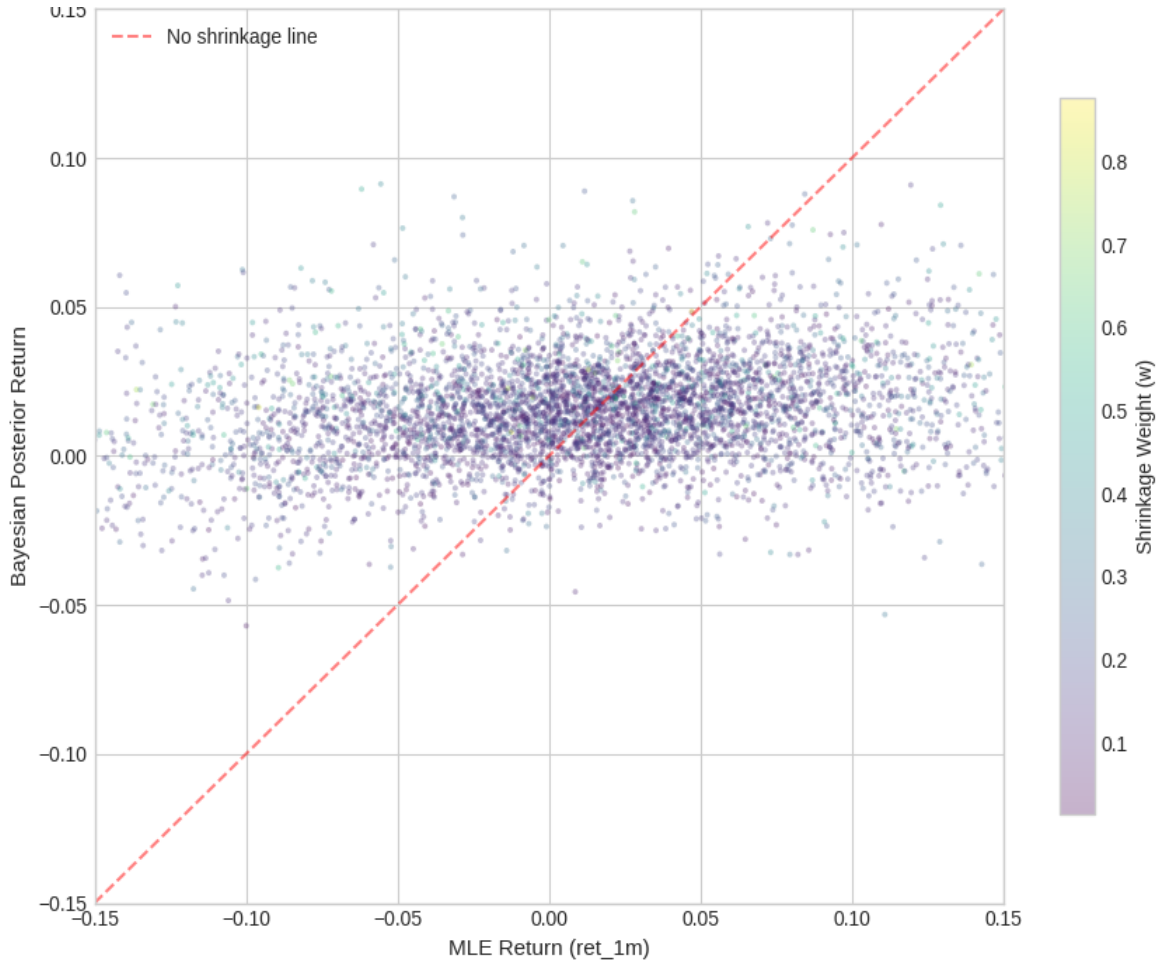

Figure A10 plots raw 12-month rolling returns against their Bayesian-shrunk equivalents, coloured by shrinkage weight, where points on the diagonal represent zero shrinkage and points pulled toward the centre reflect strong regularisation toward the sector mean, with higher shrinkage weights (lighter colour) corresponding to stocks where the prior dominates the sample estimate, stabilising the feature space and reducing the influence of outlier months on model training.

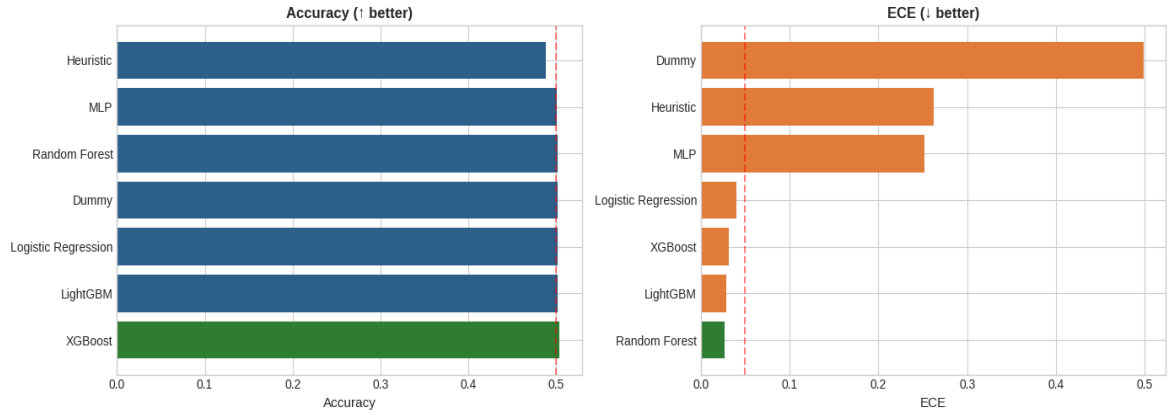

Figure A11 presents side-by-side bar charts of ECE (left) and accuracy (right) for all six models, where the visual contrast between Random Forest's low ECE bar and MLP's high ECE bar illustrates the calibration gap that motivates the entire study, while accuracy differences are much smaller, reinforcing the argument that ECE is a more informative metric than accuracy for this prediction task.